

Step 8 - Azure Setup + ADF Instance

Phase C · Canada Central · ADLS Gen2 · Cost Control · ~20 min

Goal: Stand up Azure properly: account, budget alert, resource group, ADLS Gen2 with hierarchical namespace, containers + data, ADF instance - all in Canada Central. Artifacts: screenshots + adf/README.md note.

LEARN — Money First

The free \$200 credit is once per customer, ever (tied to identity: card + phone). If you have it, the spending cap protects you. If not, pay-as-you-go is fine - this phase costs almost nothing:

Resource	Pricing	Phase C usage	Cost
ADF instance	Free to create	1	\$0
Orchestration	~\$1 per 1,000 activity runs	~50 runs	~\$0.05
Copy data movement	~\$0.25/DIU-hour, per-minute	~50 one-minute copies of 1MB	\$1-2
ADLS storage	~\$0.02/GB/month	a few MB	~\$0.001
TOTAL			approx \$2-4 CAD

The Canada Central rule: every resource goes in Canada Central (Toronto) or Canada East (Quebec City). Canadian banks keep customer data within Canadian borders - OSFI expects it, PIPEDA favours it. Saying this in interviews signals real-world awareness.

LAB

- 8.1 — Sign up (free account if eligible, else pay-as-you-go) at azure.microsoft.com - then portal.azure.com.
- 8.2 — Budget alert FIRST: Cost Management - Budgets - Add: \$5, alerts at 50% and 80% to your email.
- 8.3 — Resource group: name rg-maplebank, region Canada Central. Your delete-everything-later container.
- 8.4 — Storage account: name maplebankdl + digits (lowercase), Canada Central, Standard, LRS. CRITICAL: Advanced tab - ENABLE HIERARCHICAL NAMESPACE - that checkbox is what makes it ADLS Gen2 rather than plain Blob. Then create containers source and bronze; upload the 4 CSVs into source.
- 8.5 — Data Factory: name adf-maplebank + digits, rg-maplebank, Canada Central, V2, 'Configure Git later' - then Launch Studio (adf.azure.com).
- 8.6 — Orient in ADF Studio: Author (pencil - pipelines/datasets), Monitor (gauge - run history), Manage (toolbox - linked services, integration runtimes, triggers). Under Manage - Integration runtimes, confirm AutoResolveIntegrationRuntime.

COMMIT + CHECKPOINT

Screenshots to adf/ on GitHub: resource group view (everything Canada Central), source container with CSVs, ADF Studio home. README note: 'ADF + ADLS Gen2 in Canada Central (data residency); \$5 budget alert.' Pass criteria: all resources in Canada Central; hierarchical namespace enabled (storage account Overview shows 'Hierarchical namespace: Enabled').

If you used a workaround account against Microsoft's one-free-account terms, treat it as disposable - keep nothing there you cannot lose. GitHub and Databricks remain your real portfolio home.

